Lab 07 Model Evaluation, ITAI 1371

Team: ThetaBetas, Brandy Gonzales

Contribution Journal

Each member of our team completed the lab independently, then shared their files on Teams so our team leader could upload them to our GitHub repository.

I imported the Titanic dataset, which we used in previous labs. The notebook file had lost its line breaks, so all the code was mashed together and cell 1 would not run. I split each command onto its own line to fix it. Next, I received a pandas error because the blanks in 'Age' were not being filled. I fixed it by adding df['Age'] = to the beginning of line 7, so the filled-in values were saved back to the column.

I completed Tasks 1-3 (the confusion matrix, the classification report, and 5-fold cross validation) and ran all cells successfully. I wrote the knowledge check answers and my reflection, then exported the notebook as a PDF and shared it with my team on Teams.

Course: ITAI 1371

For this lab, my individual contribution was completing the model evaluation tasks in the notebook and interpreting the results produced by the model. I worked with the Titanic dataset and evaluated a Logistic Regression model that was used to predict whether passengers survived.

I first worked on the confusion matrix section. I used the required code to calculate the confusion matrix and created the heatmap to visualize the results. My confusion matrix produced the values 90, 15, 20, and 54. I then interpreted these numbers by identifying the true negatives, false positives, false negatives, and true positives. I found that the model made 144 correct predictions out of 179 test cases, giving an accuracy of about 80.4%. I also explained that the model made 35 incorrect predictions.

Next, I completed the classification report section. I generated the report and reviewed the precision, recall, and F1-score for both classes. For passengers who died, the model produced a precision of 0.82, recall of 0.86, and F1-score of 0.84. For passengers who

survived, the precision was 0.78, recall was 0.73, and F1-score was 0.76. I interpreted these results and explained that the model performed better at identifying passengers who died than passengers who survived.

I also completed the cross-validation section of the notebook. I used 5-fold cross-validation to evaluate the model several times instead of depending on only one train-test split. The five scores were around 77.5% to 80.3%, and the average cross-validation score was 78.68%. The standard deviation was 0.0094. I explained that the small variation between the folds showed that the model was fairly consistent, while the average cross-validation score gave a more reliable estimate of its expected performance.

Another part of my contribution was completing the Knowledge Check questions. I explained a situation where precision would be more important, using a bank fraud detection system as an example. I also explained a situation where recall would be more important, using medical screening for a serious disease such as cancer or stroke. Finally, I explained why cross-validation is generally more trustworthy than relying on a single train-test split.

While working through the notebook, I reviewed each result instead of only running the code. I tried to understand what each metric was telling me about the model. This helped me connect accuracy, precision, recall, F1-score, the confusion matrix, and cross-validation.

My contribution to this lab was completing the required evaluation tasks, checking the outputs, interpreting the results in my own words, and answering the Knowledge Check questions. The lab helped me improve my understanding of how to evaluate a classification model more carefully instead of depending only on one accuracy score.

Here is my contribution journal.

Derrick gibbs

My contribution to Module 07 focused on completing and running the model evaluation notebook and checking the results. The notebook used the Titanic dataset and a logistic regression model to predict whether a passenger survived. I worked through the training and testing results, completed the evaluation tasks, and reviewed the output to make sure the required information appeared in the finished notebook. For the confusion matrix portion, I generated the matrix and its heatmap so the correct and incorrect predictions could be seen clearly. I reviewed the four results in the matrix: 90 correct predictions for passengers who did not survive, 54 correct predictions for passengers who survived, 15

incorrect predictions of survival, and 20 missed survivors. This helped me check that the chart labels matched the model's predictions. I also completed the classification report section. I used the report to examine precision, recall, and F1-score for both groups rather than relying only on the overall accuracy. The output showed about 80% accuracy, with the model performing better on passengers who did not survive than on passengers who did. I checked that these results were displayed in the notebook. For the cross-validation section, I ran five-fold cross-validation and reviewed the individual results, average score, and standard deviation. The average was 78.68%, with a standard deviation of 0.0094. I used those numbers to consider whether the model's performance stayed similar across different parts of the dataset. I also completed the three knowledge-check questions about precision, recall, and cross-validation using examples that helped me explain the concepts in my own words. I ran the notebook in Google Collab and exported the completed work to PDF with the code and outputs visible. My part of the project was the notebook execution, evaluation work, review of the results, and written explanations described above.

Marlon Roberts

For my contribution, I assembled the repository together and assembled everyone’s essays into one, I did the lab and found the average and the deviation, went through all the steps needed.